

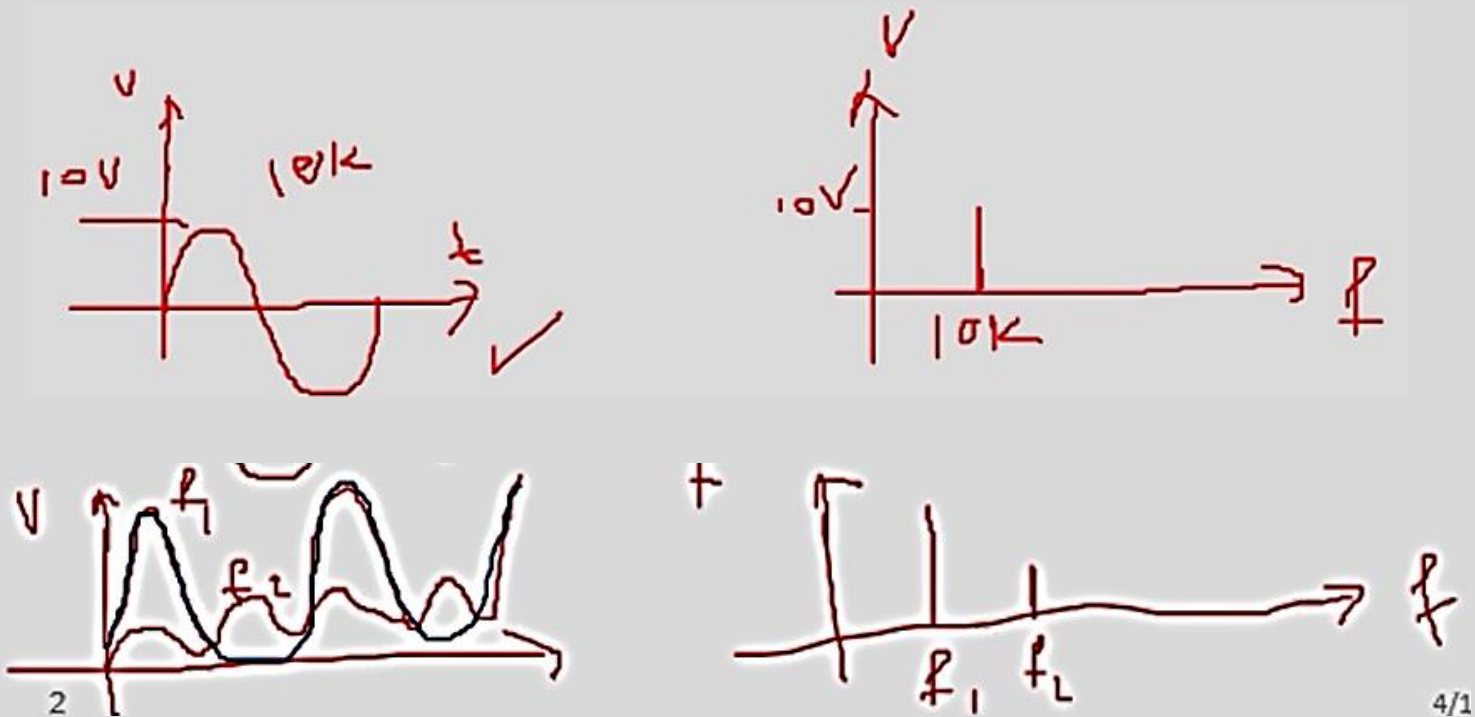
Waveform Analysis

(Ch3)

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Contents:

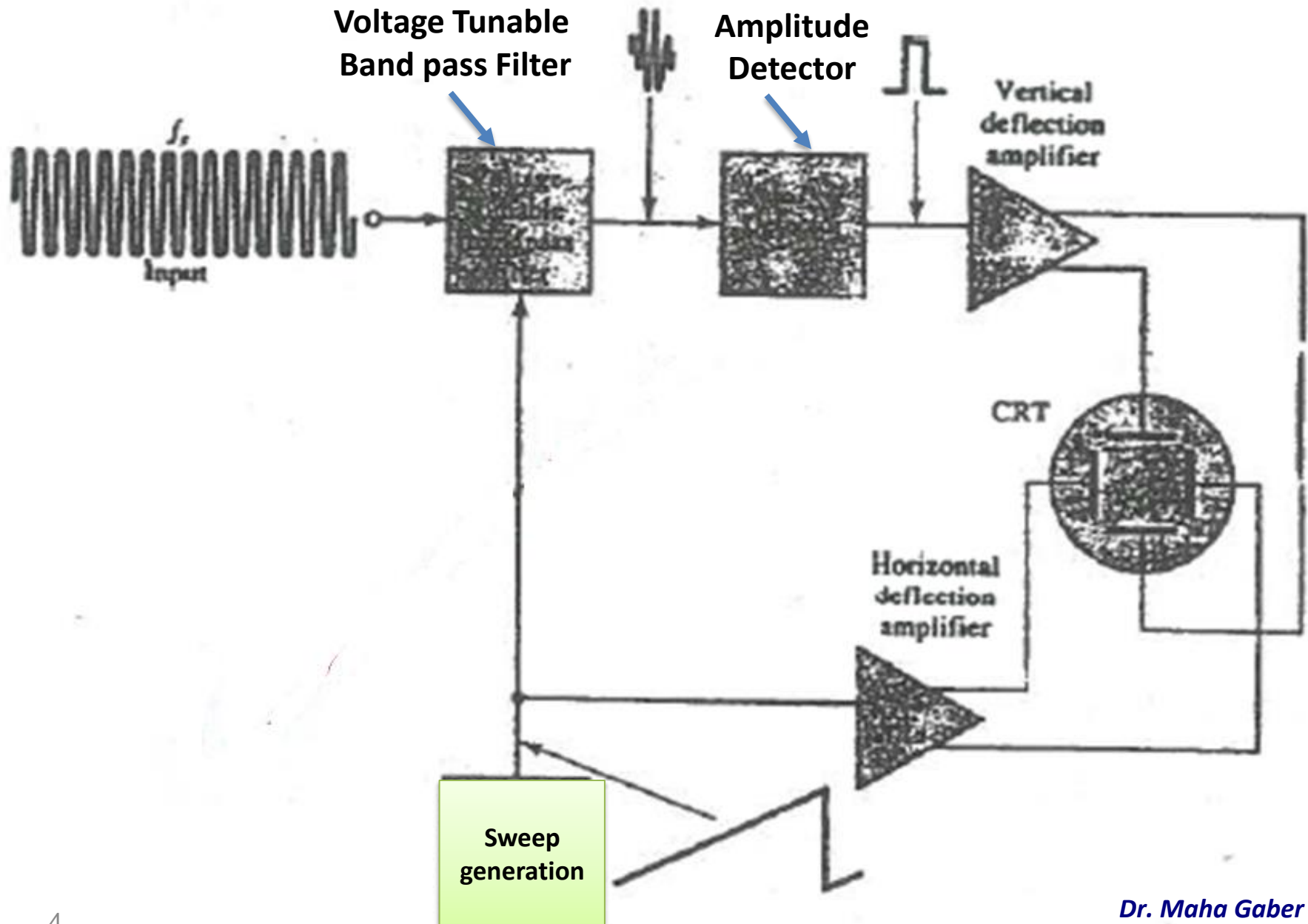
- Swept TRF Spectrum Analyzer
- Digital Spectrum Analyzer



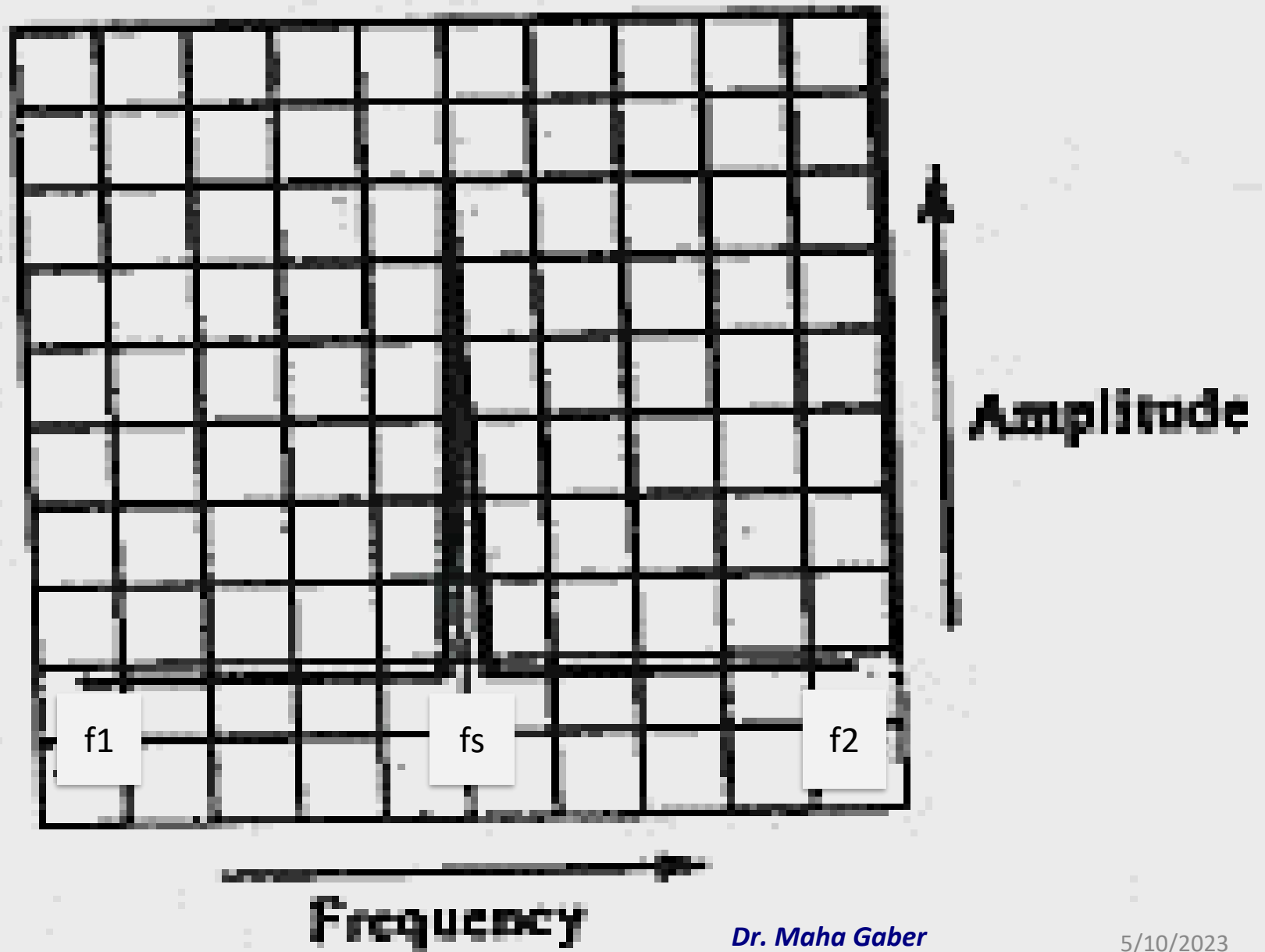
Swept TRF Spectrum Analyzer

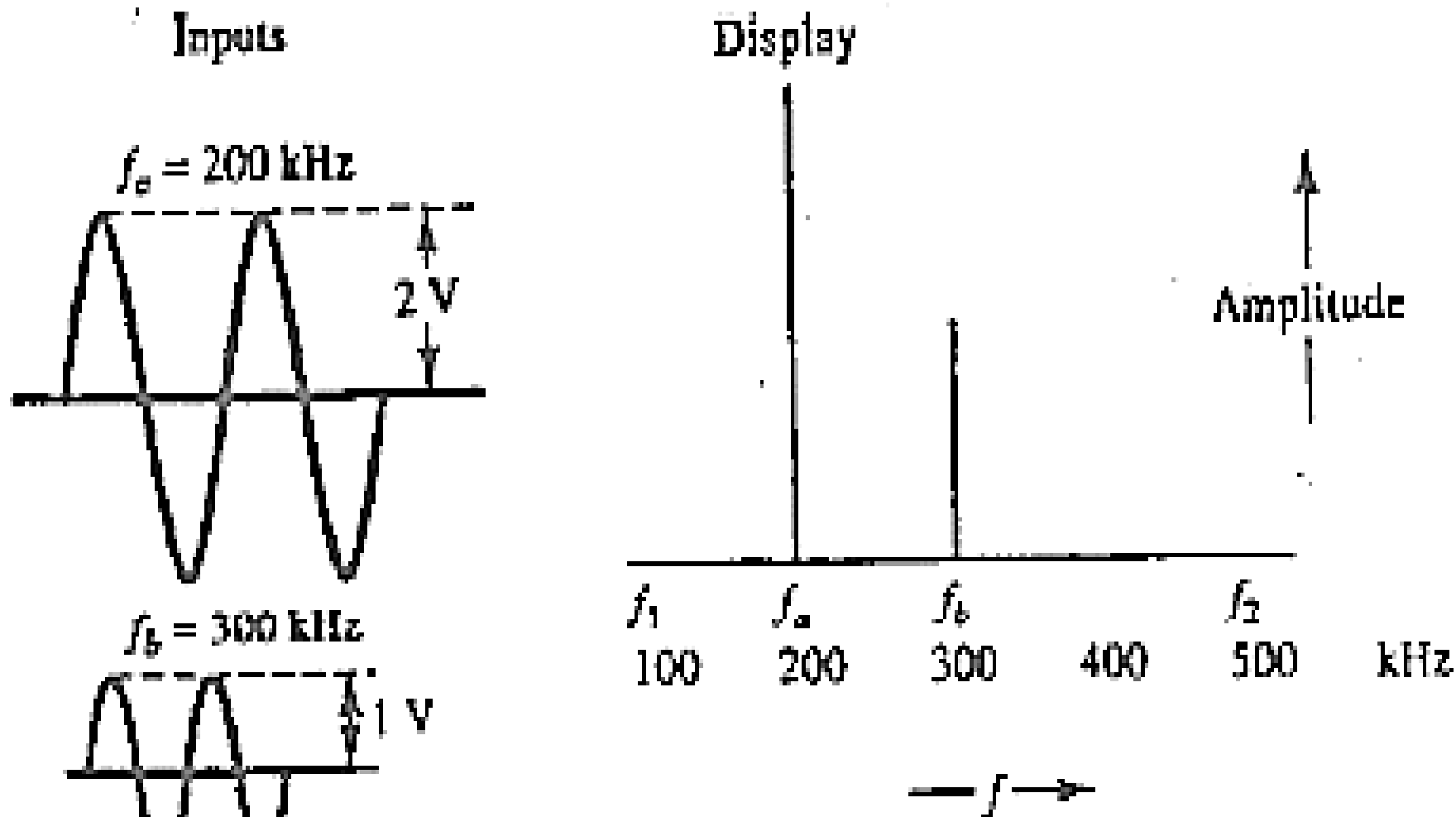
- The **spectrum analyzer separates** an **AC** signal into its various **frequency components** and **display** each **component** as a **vertical line** on a **CRT** screen.
- The **amplitude** of each **vertical line** represents the **amplitude** of each **frequency component**, and the **horizontal position** of each line defines the **frequency** of each **component**.
- The **next figure** shows a **swept TRF** (**Tuned Radio Frequency**) **spectrum analyzer**.

Swept TRF Spectrum Analyzer Block diagram

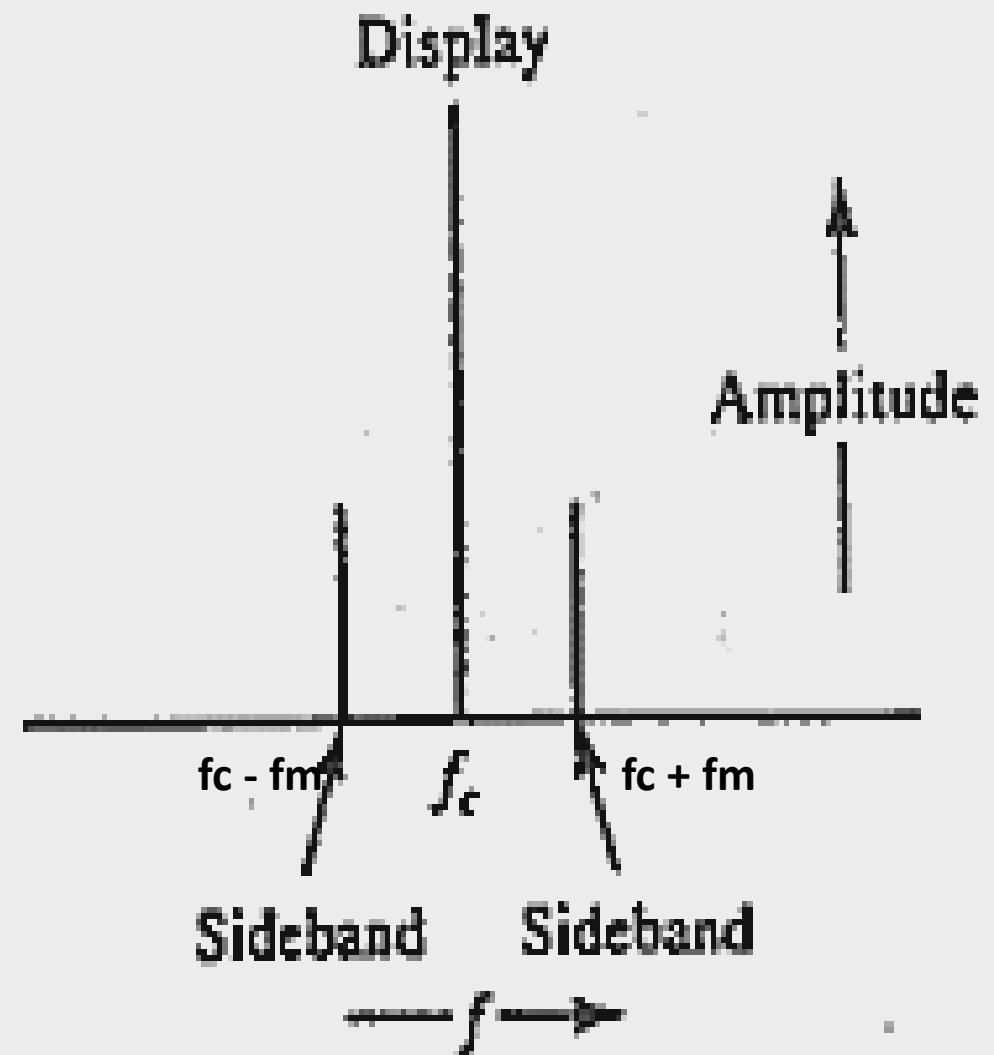
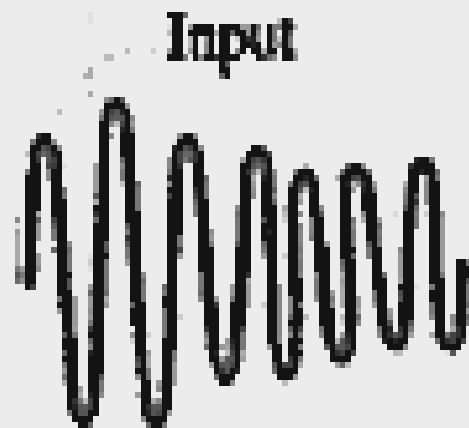


- The **sweep generator** output (**linear ramp**) is applied to a **voltage tunable bandpass filter**, the **center frequency** of the **passband** is **swept** from **min** frequency (**f_1**) to **max** frequency (**f_2**), as the **ramp voltage** sweeps from **min** to **max amplitude**.
- When an **input signal** with frequency **f_s** **passes** from the filter, it is **converted** to a **DC voltage level** by the **amplitude detector**, and applied to the **vertical** deflection amplifier of the **CRT**.
- The **next figures** illustrates **spectrum analyzer frequency domain display** by **various** input waveforms.

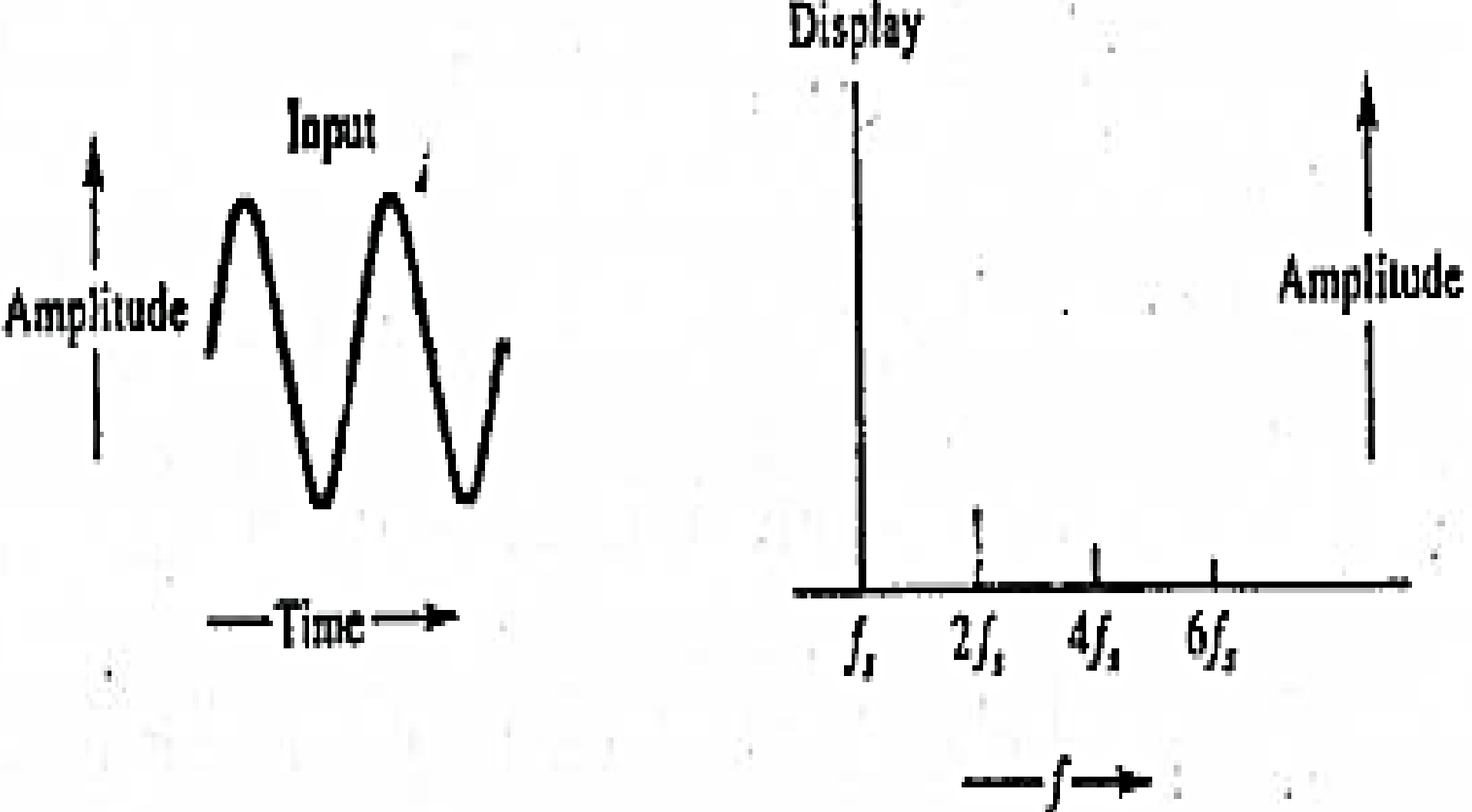




(a) Two different input signals applied simultaneously



- (b) Amplitude-modulated waveform is separated into its carrier frequency component and two sidebands.



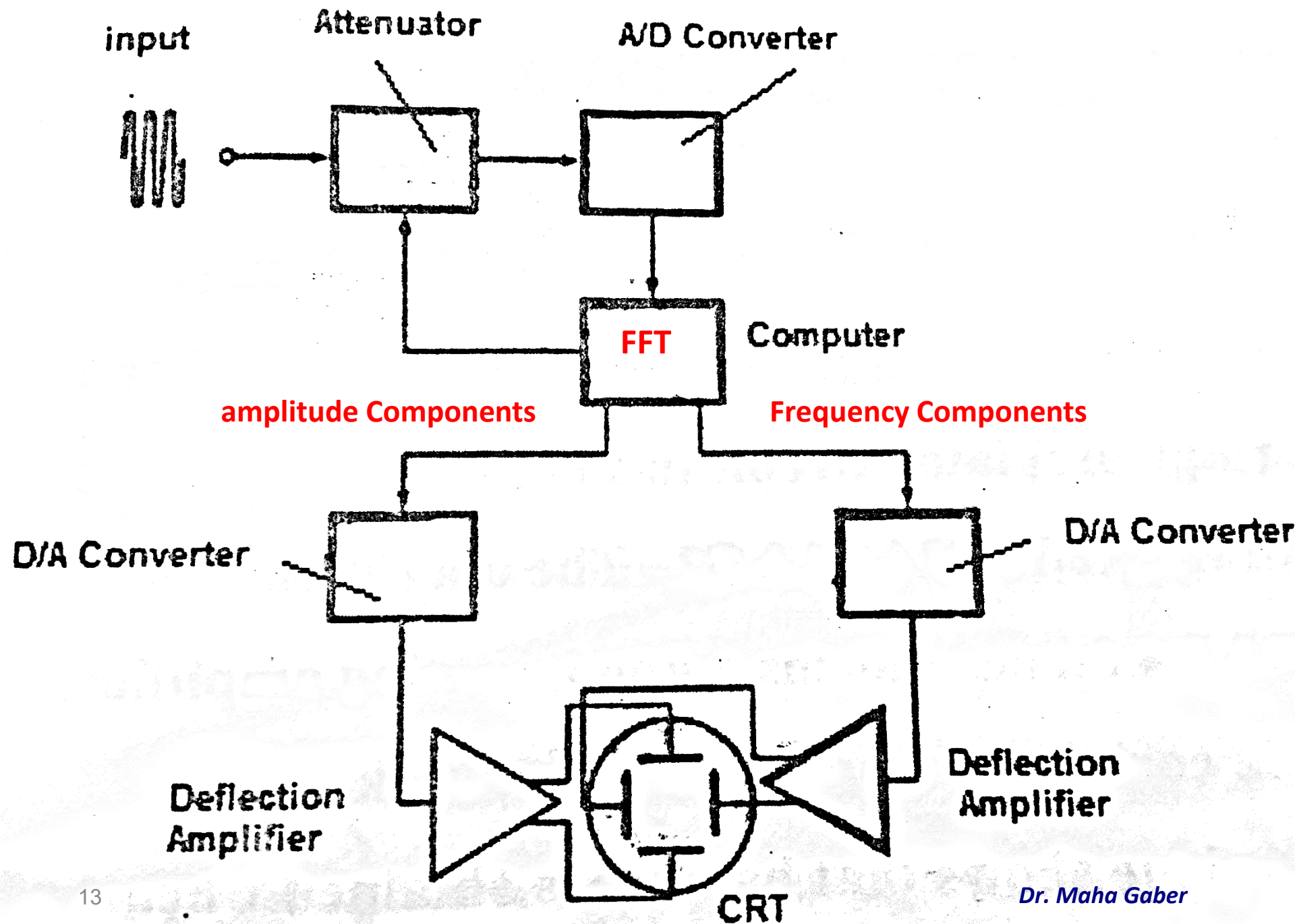
(c) An apparently perfect sine wave input might be shown to have harmonic distortion.

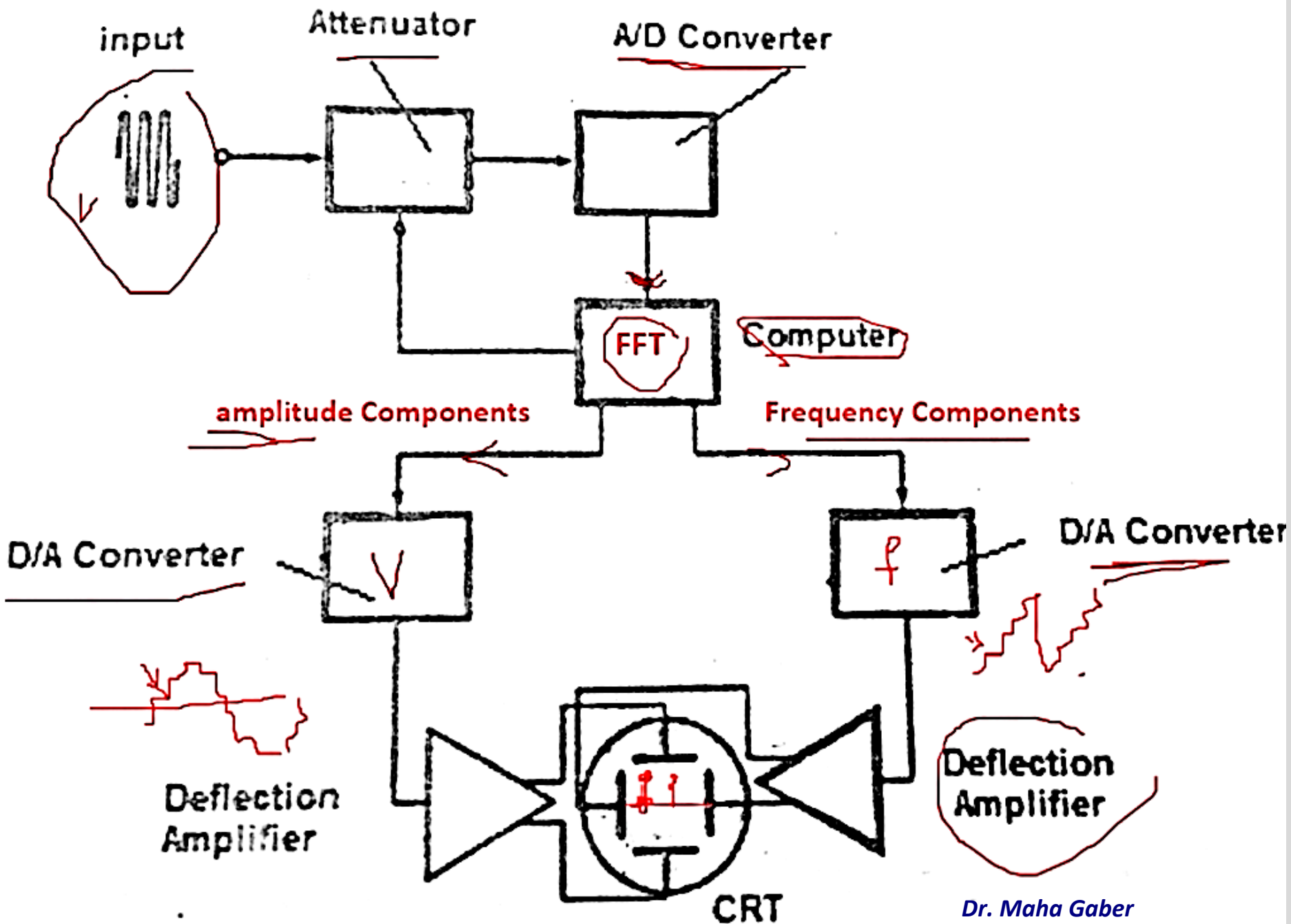
Digital (Fourier) Spectrum Analyzer

FFT analyzer

- In **Digital Spectrum Analyzer**, the **waveform** to be analyzed is sampled, and the samples are digitized and fed to a **computer**.
- The **computer** is **programmed** to determine the **waveform equation** from the **samples**, and analyze the **equation** to calculate the component waves.
- The **component waveform** are **stored** in the computer **memory**, then, when **retrieved** from **memory**, they are **converted back** to **analog** form for **display** on a **CRT screen**, as amplitude versus frequency.

- The **computer** can also **assess** the **peak** values, **rms** values, and **phase** of the **components waves**.
- The **sampling** and **conversion** techniques used in the **digital spectrum analyzer** are the same as those used in a **digital oscilloscope**.
- The next figure shows the basic block diagram for **digital** (or **Fourier**) **spectrum analyzer**.





- The **attenuator** operates as a **switch** that applies the input waveform to the **ADC** for a **short fixed** amount of **time (time window)**
- To avoid waveform **distortion**, the **time window** must have **smooth rise** and **fall** edges, this type of **windows (bell shape window)**, with a **slow** open/close characteristics, are called **Hann window**.
- The **Hann window** opens **slowly** to a peak level for a **short time**, then closes **slowly** in the same way.

- The **waveform** applied to the **computer** during this **window time(waveform passes as parts)** is assumed to be **repetitive**.
- The **algorithm** used by the **computer** is the **Fast Fourier Transform FFT**, therefore this type of instrument is also known as an **FFT analyzer**
- **Fourier analyzer** are essentially **low-frequency instrument**, mostly used for investigating the characteristics of **audio** amplifier, filters, loudspeakers, and **mechanical** vibration analysis, after converting to **electrical signals** from **transducer**.